DIVISION

I CAN SOLVE DIVISION PROBLEMS IN MANY WAYS.

DIVIDEND

In a division problem, the dividend is the number being divided.

DIVIDEND

$$12 \div 3 = 4$$

DIVISOR

In a division problem, the divisor is the number you are dividing the dividend by .

DIVISOR

$$12 \div 3 = 4$$

QUOTIENT

In a division problem, the quotient is the result, or the answer to the problem.

QUOTIENT

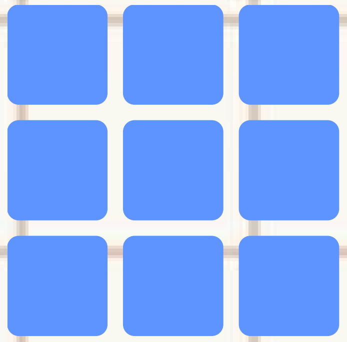
$$12 \div 3 = 4$$

QUOTIENT

$$12 \div 3 = 4$$

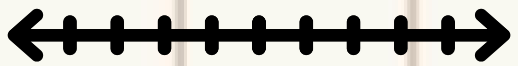
WAYS TO DIVIDE

CREATE AN ARRAY



SPLIT INTO EQUAL GROUPS

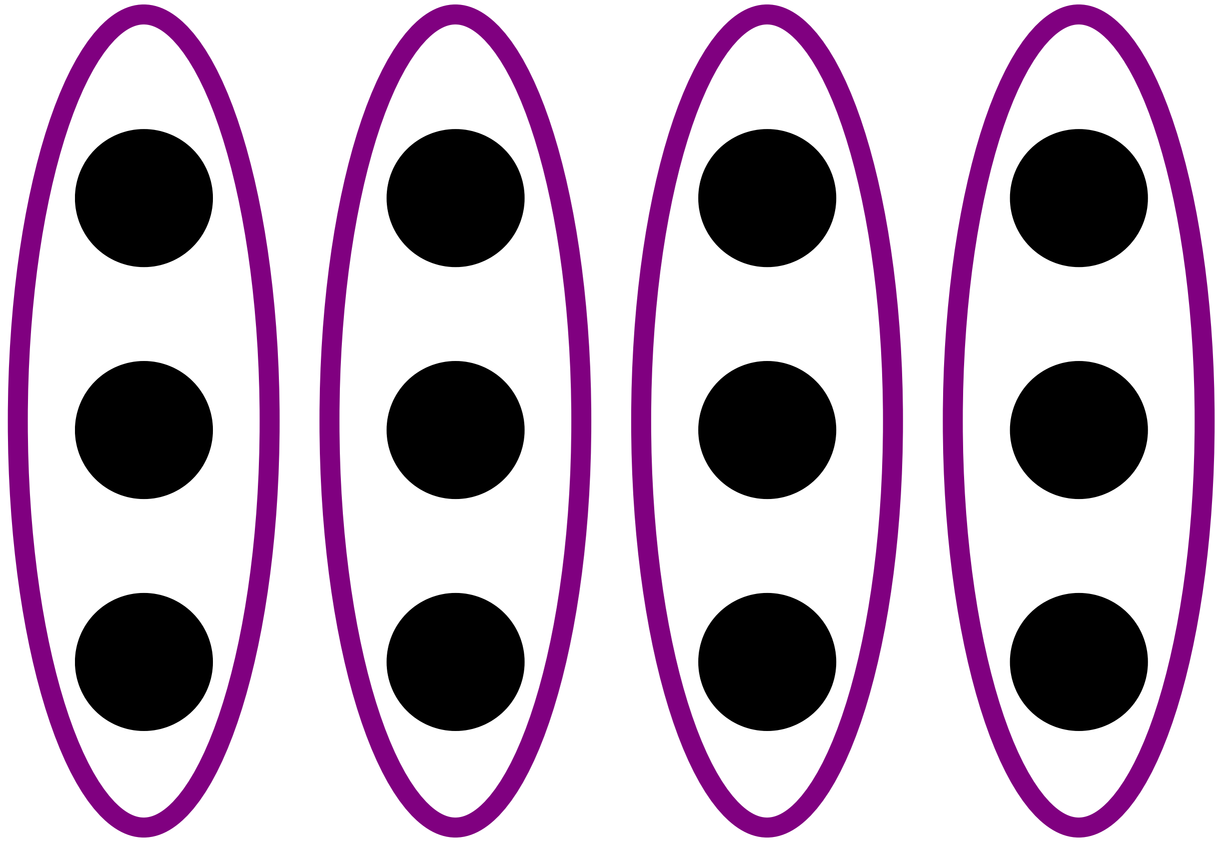
USE A NUMBERLINE



REPEATED SUBTRACTION

ARRAY

$12 \div 3 = 4$



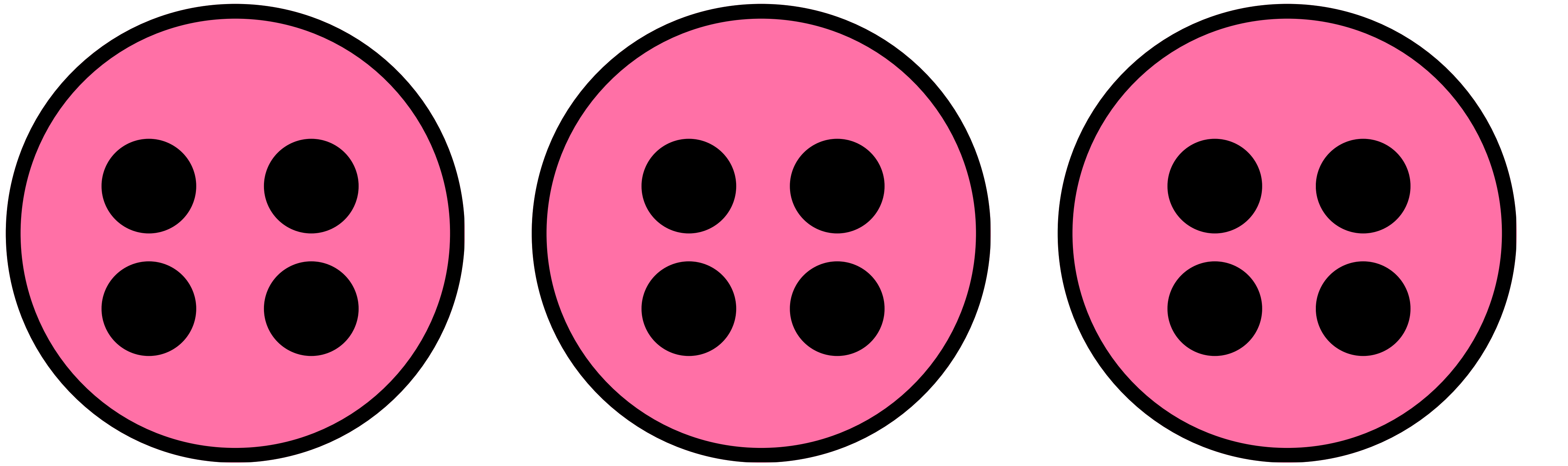
ARRAY

$21 \div 3 =$

PRACTICE

Equal Groups

$12 \div 3 = 4$



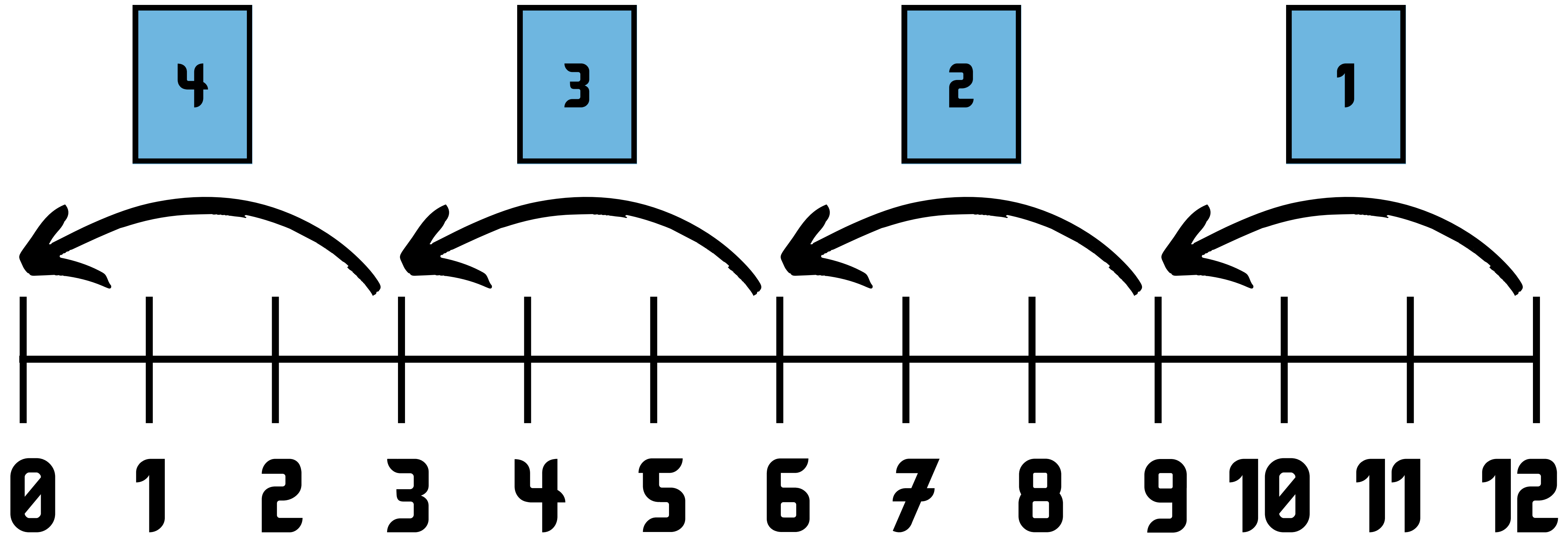
EQUAL GROUPS

$16 \div 4 =$

PRACTICE

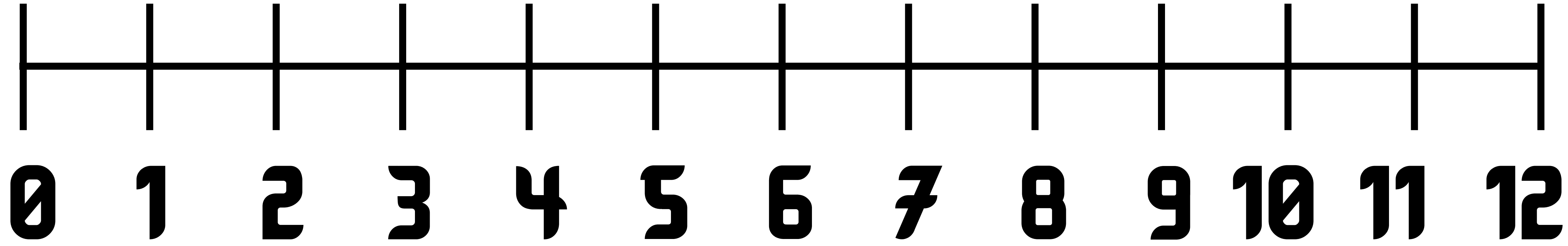
Number Line

$12 \div 3 = 4$



NUMBERLINE

$10 \div 2$



PRACTICE

Repeated Subtraction

$12 \div 3 = 4$

1	2	3	4
12	9	6	3
-3	-3	-3	-3
9	6	3	0

SUBTRACTION

$15 \div 5$

PRACTICE

Relating Multiplication & Division

Terms in Multiplication

The numbers which are multiplied are called the **factors**.

The number that results from multiplying the two **factors** is called the **product**.

$$\underbrace{5 \times 4}_{\text{Factors}} = \underbrace{20}_{\text{Product}}$$

Terms in Division

$$20 \div 4 = 5$$

Dividend Divisor Quotient

The number that is divided is called the **dividend.**

The number that divides the number is called the **divisor.**

The answer we get through the process of division is called the **quotient.**

Relating Multiplication & Division

Division is the **inverse**, or "opposite" of multiplication.

This means, if in multiplication we find the **product** of the two **factors**, in division, we find the **missing factor** if the other **factor** and the **product** are known.

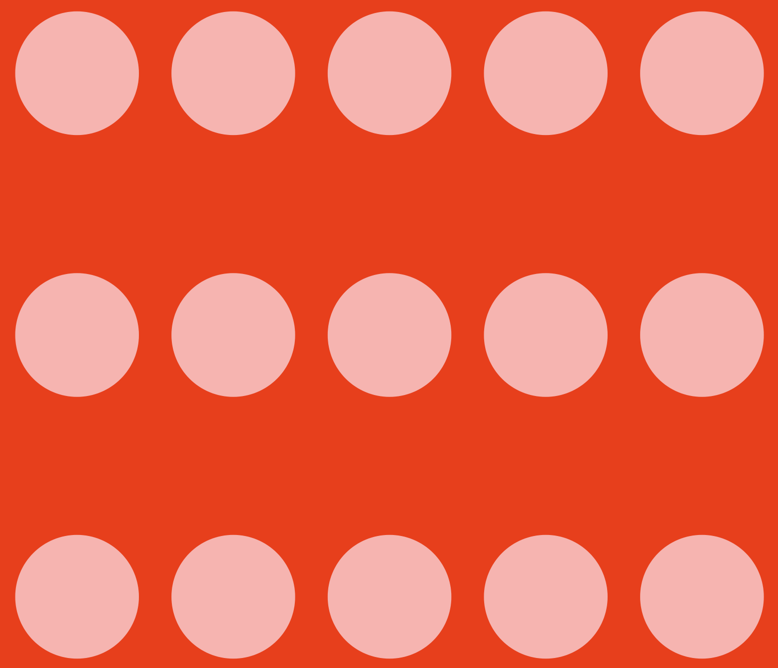
Multiply with Equal Groups

number of groups

number in all

$$3 \times 5 = 15$$

number in each group



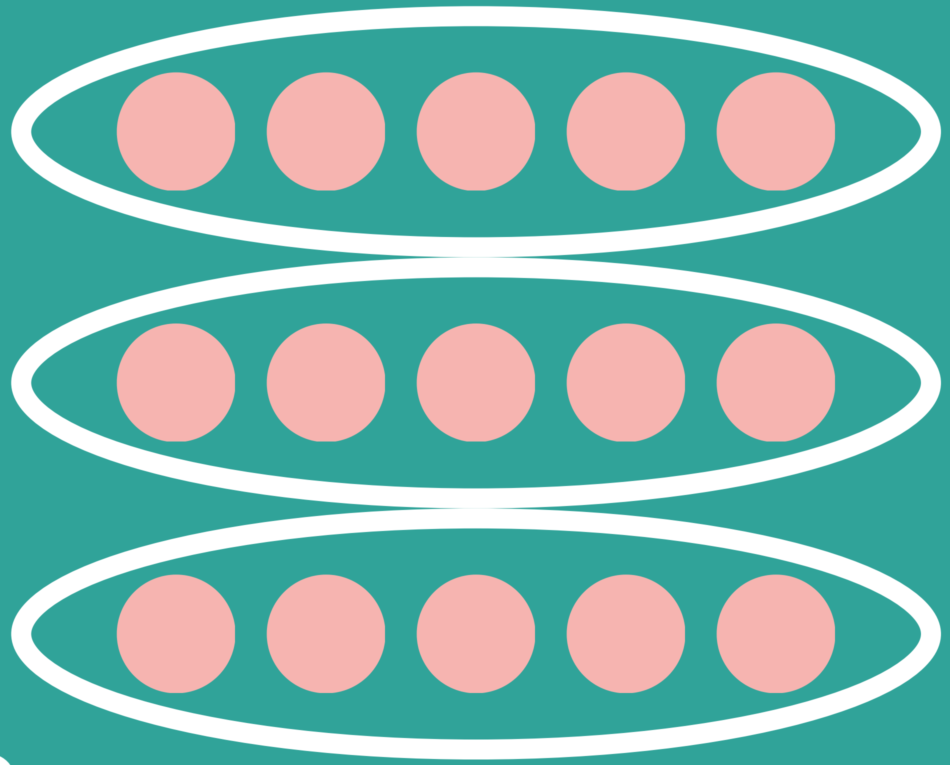
Divide with Equal Groups

number of groups

$$15 \div 3 = 5$$

number in all

number in each group



Let's Practice!

Put your multiplication and division skills to the test!

Using Multiplication

Use the multiplication equation to help answer the division equation.

If

$4 \times 6 = 24$

Then

$24 \div 4 = \boxed{?}$

If

$5 \times 7 = 35$

Then

$35 \div 7 = \boxed{?}$

Answers: Using Multiplication

Use the multiplication equation to help answer the division equation.

If

$4 \times 6 = 24$

Then

$24 \div 4 = \boxed{6}$

If

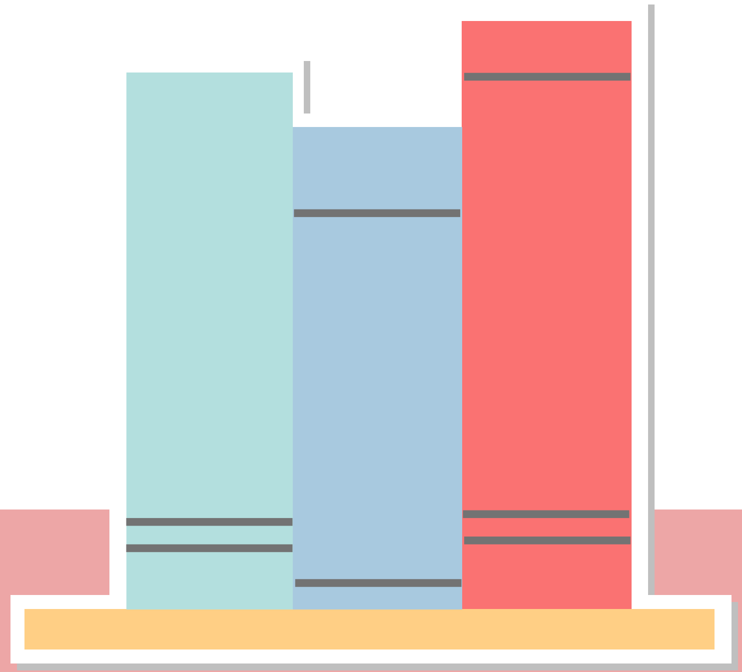
$5 \times 7 = 35$

Then

$35 \div 7 = \boxed{5}$

Divisibility

Divisibility means that you can divide a whole number by another whole number evenly or without a remainder.



Divisibility Rules for 2

A number is divisible by 2 if its last digit is 0, 2, 4, 6, or 8. All even numbers are divisible by 2.

45 62<u>8</u>

Divisibility Rules for 3

A number is divisible by 3 if the sum of all the digits is divisible by 3 or a multiple of 3.

15 324 = 1 + 5 + 3 + 2 + 4 = 15

Divisibility Rules for 4

A number is divisible by 4 if its last two digits form a number that is divisible by 4 or if its last two digits are both zeros.

53 416

87 600

Divisibility Rules for 5

A number is divisible by 5 if its last digit is 0 or 5.

63 745 ___

Divisibility Rules for 6

A number is divisible by 6 if it is divisible by both 2 and 3.

✓ Divisible by 2

✓ Divisible by 3

Divisibility Rules for 8

A number is divisible by 8 if its last three digits form a number that is divisible by 8 or if its last three digits are all zeros.

53 208

24 000

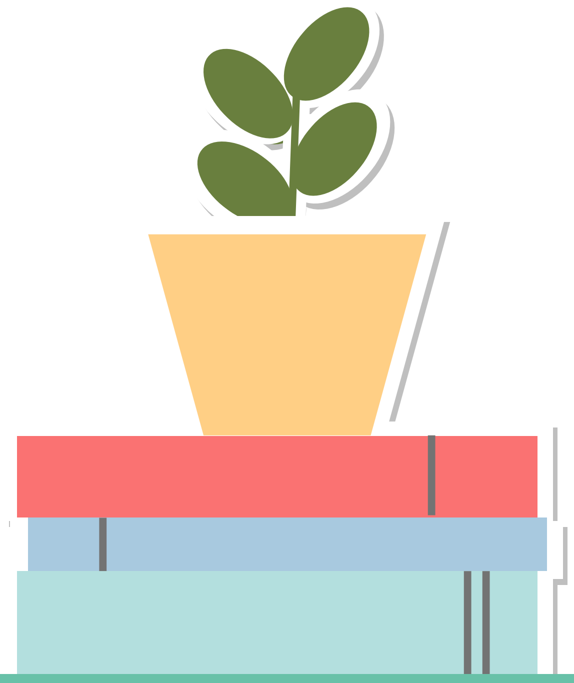
Divisibility Rules for 9

A number is divisible by 9 if the sum of all the digits is divisible by 9 or a multiple of 9.

$$27\,324 = 2 + 7 + 3 + 2 + 4 = 18$$

Divisibility Rules for 10

A number is divisible by 10 if its last digit is 0.



32 640

Thank You!